Exercise 5 • Corporate Income Statement & Trade Discounts

Corporate Income Statement, Tiered Discounts, Taxes & Dividends

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Prepare the Income Statement of a company with the following information from year 2017:

- *Gross sales: 2.002 million €. Trade policy gives 3% discount to Type A customers (annual purchases > 120 million €). Type A customers account for 40% of sales. Total sales = Gross sales - Discount.*
- *Purchases of raw materials: 962 million €. Assume no change in stocks.*
- *Workforce: 7.800 employees, average salary 30.000 €/year, Social Security spending 30% of salary.*
- *Other operating expenses: 457 million €.*
- *In recent years invested 1.200 million € in fixed assets (lifespan 10 years, linear depreciation, no residual value).*
- *International loan obtained in 2015 of 225 million € at 6% interest along the whole year, principal repayable in 2020.*
- *Corporate tax rate: 30%. Agreed dividend: 50%.*

EXAM DAY STRATEGY: DISCOUNT ELIGIBILITY, SOCIAL SECURITY & DIVIDEND FLOW

- **Commercial Discount Eligibility:**

$$\text{Type A Customer Purchases} = 40\% \times 2,002.00 \text{ M€} = \mathbf{800.80} \text{ M€}$$

The 120 M€ threshold is a per-customer criterion that defines who is Type A; the statement already gives their aggregate share (40% of sales), so no test of 800.80 against 120 is needed. The 3% discount applies to that aggregate:

$$\text{Discount} = 3\% \times 800.80 = \mathbf{24.024} \text{ M€} \implies \text{Net Sales} = 2,002.00 - 24.024 = \mathbf{1,977.976} \text{ M€}$$

- **Workforce Labor Cost (Gross Salary + Employer Social Security):**

$$\text{Base Wages} = 7,800 \times 30,000 \text{ €} = 234.00 \text{ M€}$$

$$\text{Employer Social Security} = 30\% \times 234.00 = 70.20 \text{ M€} \implies \text{Total Labor Expense} = \mathbf{304.20} \text{ M€}$$

- **Exam Trap 1 – Discount Scope:** Never apply the 3% discount to the full 2,002 M€! The prompt states only Type A customers (40%) qualify. Applying the discount to total sales deducts 60.06 M€ instead of 24.024 M€, distorting all downstream figures.
- **Exam Trap 2 – Principal Repayment vs Interest:** The loan principal (225 M€) matures in 2020. **Principal repayments are Balance Sheet cash/debt movements, NEVER Income Statement expenses!** Only the annual interest ($6\% \times 225 = 13.50$ M€) is recorded in the P&L.
- **Exam Trap 3 – Dividend Payout Base:** Apply the 50% dividend rate to **Net Income (profit after corporate taxes)**, never to EBIT or EBT. Dividends represent profit distribution to shareholders after all statutory obligations are fulfilled.

Phase 1: Step-by-Step Numerical Deductions

1. **Gross Sales, Commercial Discount & Net Sales:**

$$\text{Type A Volume} = 0.40 \times 2,002.00 = 800.80 \text{ M€} \quad \text{(the 120 M€ threshold is per customer and only defines the Type A class)}$$

$$\text{Commercial Discount} = 0.03 \times 800.80 = \mathbf{24.024} \text{ M€}$$

$$\text{Net Sales Turnover} = 2,002.00 - 24.024 = \mathbf{1,977.976}\ \text{M€}$$

2. **Cost of Goods Sold (COGS):** Because raw material stocks did not change ($\Delta\text{Stock} = 0$):

$$\text{COGS} = \text{Purchases of Raw Materials} = \mathbf{962.000}\ \text{M€}$$

$$\text{Gross Profit} = \text{Net Sales} - \text{COGS} = 1,977.976 - 962.000 = \mathbf{1,015.976}\ \text{M€}$$

3. **Operating Expenses (Gastos de Explotación):**

$$\begin{aligned}
\text{Base Salaries (7,800 workers } \times 30,000\ €) &= 234.000\ \text{M€} \\
\text{Employer Social Security (30\% on salaries)} &= +70.200\ \text{M€} \\
\text{Total Personnel Expense} &= \mathbf{304.200}\ \text{M€} \\
\text{Other Operating Expenses} &= \mathbf{457.000}\ \text{M€} \\
\text{Linear Depreciation } \left(\frac{1,200.00}{10\ \text{yrs}}\right) &= \mathbf{120.000}\ \text{M€} \\
\hline
\text{Total Operating Expenses} &= 304.200 + 457.000 + 120.000 = \mathbf{881.200}\ \text{M€}
\end{aligned}$$

4. **Operating Income / EBIT (Resultado de Explotación):**

$$\text{EBIT} = \text{Gross Profit} - \text{Operating Expenses} = 1,015.976 - 881.200 = \mathbf{134.776}\ \text{M€}$$

5. **Financial Expenses (Intereses de la Deuda):** Outstanding loan balance in 2017 is 225 M€ at 6% per annum:

$$\text{Financial Interest Expense} = 0.06 \times 225.00 = \mathbf{13.500}\ \text{M€}$$

6. **Earnings Before Taxes / EBT (Resultado Antes de Impuestos):**

$$\text{EBT} = \text{EBIT} - \text{Interest Expense} = 134.776 - 13.500 = \mathbf{121.276}\ \text{M€}$$

7. **Corporate Income Taxes (30%) & Net Income:**

$$\text{Corporate Tax} = 30\% \times \text{EBT} = 0.30 \times 121.276 = \mathbf{36.3828}\ \text{M€}$$

$$\text{Net Income (Net Profit)} = 121.276 - 36.3828 = \mathbf{84.8932}\ \text{M€}$$

8. **Profit Appropriation (Dividends vs Retained Reserves):**

$$\text{Dividends to Shareholders (50\% Payout)} = 0.50 \times 84.8932 = \mathbf{42.4466}\ \text{M€}$$

$$\text{Retained Earnings to Equity Reserves (50\% Retention)} = 0.50 \times 84.8932 = \mathbf{42.4466}\ \text{M€}$$

Phase 2: Formal Corporate Income Statement (Year 2017)

INCOME STATEMENT – FISCAL YEAR 2017	Detail (M€)	Amount (M€)
Gross Sales Turnover	2,002.000	
(-) Commercial Discounts (3% on Type A customers)	(24.024)	
= **NET SALES REVENUE**		**1,977.976**
(-) Cost of Goods Sold (Raw materials purchases)		(962.000)
= **GROSS PROFIT**		**1,015.976**
(-) Operating Expenses		(881.200)
• Personnel Cost (Wages 234 M€ + 30% SS 70.2 M€)	(304.200)	
• Other Operating Expenses	(457.000)	
• Linear Depreciation of Fixed Assets (10% of 1,200 M€)	(120.000)	
= **OPERATING INCOME / EBIT**		**134.776**
(-) Financial Interest Expenses (6% of 225 M€ loan)		(13.500)
= **EARNINGS BEFORE TAXES / EBT**		**121.276**
(-) Corporate Income Tax (30%)		(36.383)
NET INCOME FOR THE YEAR		**84.893**
• Dividends to be distributed (50% payout)		42.447
• Retained Earnings transferred to Equity reserves (50%)		42.447

FINANCIAL RATIOS & PAYOUT SUMMARY

Operating Return on Sales (ROS): $\frac{\text{EBIT}}{\text{Net Sales}} = \frac{134.776}{1{,}977.976} = \mathbf{6.81\%}$

Net Profit Margin: $\frac{\text{Net Income}}{\text{Net Sales}} = \frac{84.893}{1{,}977.976} = \mathbf{4.29\%}$

Interest Coverage Ratio (TIE): $\frac{\text{EBIT}}{\text{Interest}} = \frac{134.776}{13.500} = \mathbf{9.98}\times$ (Strong solvency: operating profit covers debt interest nearly 10 times over).

Study notes written from the official course material (T3_Financial Statements_EXERCISES.pdf; Topic 3 Financial Management (I).pdf). Online version: subjects/business-management/problemas/topic-3-financial-statements.html